

C# OOP Assignment

OOP 02 – Smart Delivery Management System

Part 01 : Theoretical Questions

Question 1

Answer the following questions:

- a) What is the difference between a **class** and a **struct**?

الـ Class تعتبر **Reference Type** أما الـ Struct فهي **Value Type**

- b) Why are classes more suitable than structs for large applications?

لأنها بتدعم كل مفاهيم الـ OOP بدون استثناء

Question 2

Consider the following code:

```
public class Shipment
{
    public string TrackingCode { get; set; }
}

public class ExpressShipment : Shipment
{
    public decimal ExtraFee { get; set; }
}
```

- a) Which class is the parent class? **Shipment**
- b) Which class is the child class? **ExpressShipment**
- c) What members are inherited by **ExpressShipment**? **TrackingCode**
- d) Why is inheritance better than duplicating the same code in multiple classes?

لأن الـ Inheritance بتخلينا نعيد استخدام الكود المشترك بدل ما نكتبه مرة ثانية في كل Class

ROUTE